

SUKKUR ROHRI, Pakistan

WMO: 417250

Lat: 27.667N

Lon: 68.900E

Elev: 68

StdP: 100.51

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	9.0	-1.9	3.3	20.8	0.7	4.0	19.4	4.1	16.3	3.4	16.5	0.4	50	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.0	45.1	24.5	43.8	24.8	42.5	25.4	29.2	38.6	28.8	38.1	28.5	37.7	1.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.1	34.0	26.6	22.3	34.2	26.1	21.7	34.3	97.1	39.0	94.9	38.5	93.1	38.2	34.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.4	3.5	3.0	4.9	47.9	2.7	0.9	2.9	48.6	1.3	49.1	-0.2	49.6	-2.2	50.3		
(5)			3.7	31.4	1.9	1.2	2.4	32.2	1.3	32.9	0.2	33.6	-1.1	34.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	15.7	19.0	24.9	31.7	36.1	36.7	35.6	33.6	32.6	29.1	22.9	17.3
	DBStd	7.73	2.13	2.84	3.67	3.03	2.17	1.88	1.58	1.57	1.55	2.23	2.62	2.48
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	161	85	22	2	0	0	0	0	0	0	0	1	51
	CDD10.0	6558	177	253	463	651	809	802	793	731	677	593	386	225
	CDD18.3	3677	3	42	207	401	551	552	535	473	427	334	136	17
	CDH23.3	55336	34	343	2173	5872	9193	9378	8894	7459	6507	4173	1161	148
CDH26.7	36947	1	84	1065	3791	6731	6983	6415	4988	4128	2328	413	19	
Wind	WSAvg	1.0	0.6	0.9	0.9	1.0	1.1	1.6	1.6	1.4	1.1	0.6	0.6	0.5
Precipitation	PrecAvg	116	5	7	5	7	4	9	38	27	10	3	0	5
	PrecMax	458	48	36	23	50	34	39	211	144	186	58	5	112
	PrecMin	11	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	96	11	10	7	12	8	12	56	41	37	11	1	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	31.5	39.9	44.5	47.1	46.9	43.8	40.9	40.9	39.2	34.4	29.1
		MCWB	15.9	18.7	21.4	22.5	23.6	24.6	28.1	27.7	26.3	24.3	20.7	18.9
	2%	DB	24.3	29.3	36.9	42.6	45.4	45.1	42.5	39.9	39.5	37.8	32.5	26.6
		MCWB	15.4	18.8	20.7	22.1	23.9	25.3	28.1	27.6	26.4	24.0	20.3	17.1
	5%	DB	22.7	27.2	34.5	40.9	44.1	43.6	41.4	39.0	38.4	36.5	30.9	24.9
		MCWB	14.5	17.7	20.4	22.1	23.7	26.0	28.1	27.5	26.3	23.5	20.1	16.1
	10%	DB	21.3	25.3	32.3	39.1	42.7	42.3	40.4	38.1	37.4	35.2	29.2	23.3
		MCWB	13.9	16.6	20.1	21.8	24.0	26.6	28.0	27.6	26.2	22.9	19.2	15.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	21.4	23.6	25.4	28.3	29.8	30.0	29.6	29.1	27.3	22.9	19.5
		MCDB	23.3	28.1	34.7	38.6	39.7	39.9	40.4	36.4	36.8	34.9	30.8	26.2
	2%	WB	16.4	20.2	22.4	24.4	27.5	28.9	29.4	28.9	28.4	26.4	22.0	18.4
		MCDB	22.4	26.7	32.9	37.5	39.4	39.8	39.5	36.4	36.0	34.1	29.7	24.6
	5%	WB	15.6	19.0	21.6	23.7	26.7	28.4	28.9	28.6	27.8	25.6	21.0	17.4
		MCDB	21.4	26.0	31.3	36.6	38.8	39.1	38.9	36.2	35.3	33.3	28.7	23.2
	10%	WB	14.7	17.8	20.8	23.0	25.9	28.0	28.6	28.2	27.4	24.7	20.2	16.3
		MCDB	20.1	23.9	30.1	35.8	38.1	38.5	38.4	35.9	34.9	32.4	27.3	22.0
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.0	12.5	13.6	13.5	12.0	10.3	9.1	9.7	12.1	12.3	11.8
		MCDBR	13.3	14.2	15.5	16.1	15.0	13.7	11.6	10.5	11.2	13.0	13.4	13.5
		MCWBR	6.3	6.0	5.2	4.8	4.6	4.0	3.3	3.3	3.3	4.0	4.8	6.1
	5% WB	MCDBR	12.0	12.5	13.6	14.7	13.5	11.7	10.6	9.1	9.3	11.0	12.1	12.2
		MCWBR	6.5	5.9	5.2	5.3	5.3	4.0	3.7	3.5	3.4	4.2	5.0	6.6
Clear-Sky Solar Irradiance	taub	0.487	0.493	0.537	0.606	0.645	0.721	0.799	0.723	0.598	0.553	0.573	0.525	
	taud	1.864	1.891	1.801	1.694	1.648	1.560	1.481	1.598	1.787	1.780	1.700	1.783	
	Ebn at Noon	727	769	762	725	697	642	593	634	704	703	637	663	
	Edh at Noon	175	184	213	244	256	278	300	265	213	203	205	183	
All-Sky Solar Radiation	RadAvg	3.83	4.70	5.87	6.75	7.22	6.91	6.41	6.18	6.03	5.31	4.16	3.62	
	RadStd	0.21	0.30	0.26	0.17	0.13	0.20	0.27	0.31	0.26	0.20	0.18	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page